

Kunal Maheshwari

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SUMMARY

OPT-EAD holder with full-time work authorization, bringing over 4 years of professional experience as a Software Developer and Data Engineer. Expertise in building scalable microservices and distributed data pipelines using Java, Python, Spark, SQL, AWS, and Azure. Proven track record in CI/CD automation, large-scale data platforms, and privacy-preserving AI applications, seeking a Senior Software Engineer, Streaming role to leverage skills in distributed systems and infrastructure-as-code.

EDUCATION

Central Michigan University | Mount Pleasant, MI Aug 2024 - Dec 2025
Master of Science | Computer Science

Lovely Professional University | Punjab, India Aug 2016 - May 2020
Bachelor of Technology | Computer Science and Engineering

EXPERIENCE

Software Developer | NTT DATA | Bengaluru, India (Remote) August 2022 - August 2024

- Designed and optimized high-performance REST and SOAP microservices using Java 8/11, Spring Boot, and Azure Functions, handling over 50,000 daily API requests with 99.99 percent SLA uptime and leveraging distributed systems principles.
- Built and automated CI/CD pipelines using Jenkins, Maven, and Azure DevOps, enabling continuous delivery across Azure Cloud and Oracle WebLogic environments with infrastructure-as-code practices.
- Modernized legacy systems by migrating to Log4j 2.x and improving monitoring and error handling, reducing issue resolution time by 20 percent and enhancing JVM performance.
- Led a major EDX platform upgrade, streamlining application architecture and dependency management to achieve 50 percent faster delivery cycles through effective data integration strategies.
- Collaborated within Agile and Scrum teams to deliver FHIR-compliant healthcare solutions using Git, SVN, and JUnit testing, demonstrating a strong track record of delivering complex projects.

Senior Analyst | Capgemini | Mumbai, India (Remote) October 2021 - August 2022

- Engineered Java and EJB-based enterprise applications and RESTful web services for automotive clients, optimizing for scalability and performance.
- Improved system performance by 90 percent through SQL query optimization, database tuning, and modernization of legacy Java code on IBM DB2 and Oracle databases.
- Implemented CI/CD pipelines using Jenkins and GitLab with automated testing and quality gates, reducing deployment failures by 30 percent.
- Mentored junior developers through code reviews, debugging sessions, and clean coding practices, fostering a culture of continuous learning and robust data structures.

Analyst | Capgemini | Mumbai, India (Remote) July 2020 - October 2021

- Maintained high-availability enterprise systems with over 99 percent uptime through proactive monitoring and performance optimization, utilizing tools like Datadog.
- Enhanced SQL data pipelines and database operations, improving query performance by 10 percent and strengthening data integrity across PostgreSQL and other relational databases.
- Implemented comprehensive unit testing using JUnit4, achieving 99 percent code coverage and ensuring robust code quality.
- Supported release planning and sprint execution using JIRA within Agile development teams, contributing to efficient data integration workflows.

PROJECTS

NYU Taxi Data Analysis Platform (Spark, AWS, Terraform) January 2024 - May 2025

- Built a distributed Spark-based data pipeline to process and analyze more than 8 GB of NYC taxi trip data in near real time, leveraging distributed systems principles.
- Developed an interactive Flask web application for data exploration with dynamic filtering and visualization using Tableau and QuickSight, enhancing data integration insights.
- Containerized ingestion, processing, and analytics components using Docker and Kubernetes for reproducible deployments, supporting infrastructure-as-code.
- Optimized Spark memory usage and partitioning strategies, reducing overall job runtime by 40 percent and improving JVM performance.
- Automated cloud infrastructure provisioning using Terraform across AWS services including S3, RDS, ECS, Lambda, and DynamoDB, ensuring efficient resource management.

Privacy-Preserving AI Email Client (Capstone Project) (Hugging Face, T5, LoRA) January 2025 - December 2025

- Engineered a fine-tuning pipeline for T5-based transformer models to perform email summarization and classification, utilizing Hugging Face libraries.
- Applied parameter-efficient fine-tuning using LoRA to support low-resource training under privacy constraints, demonstrating expertise in machine learning applications.
- Implemented model evaluation using ROUGE metrics and integrated quantization for secure local deployment, showcasing proficiency in advanced data structures and serialization techniques.

SKILLS

- **Programming Languages:** Proficient in Java and Python with extensive SQL experience for complex data analysis and ETL pipelines. Expertise in JVM and data structures.
- **Frameworks & Libraries:** Experienced with Spring Boot, Apache Spark, PySpark, Hugging Face, T5, and LoRA for machine learning, and JUnit for test-driven development.
- **Web Services:** Skilled in designing and implementing REST API and SOAP API solutions for distributed systems.
- **Cloud Platforms:** Adept at utilizing AWS (EC2, RDS, S3, Lambda, DynamoDB), Azure (Functions, DevOps, ServiceBus), and Google Cloud Platform for scalable infrastructure-as-code deployments.
- **Databases:** Proficient with PostgreSQL, Oracle Database, IBM DB2, and RDS, ensuring robust data integration and management.
- **Tools & Technologies:** Expertise in Docker, Kubernetes, Jenkins, Git, SVN, Maven, JIRA, Confluence, Tableau, DBeaver, and Terraform for efficient CI/CD and infrastructure automation.
- **Concepts:** Strong understanding of ETL Pipelines, Data Warehousing, Data Analytics, Agile, Scrum, Test Driven Development, Object Oriented Programming, CI/CD, Microservices, Large-scale data platforms, Infrastructure automation, and Privacy-preserving LLM-based applications.
- **Data Analysis & Visualization:** Skilled in using Tableau and QuickSight for insightful data analysis and visualization, with a solid track record of deriving actionable insights.